

Finlay Cooper

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Education

Western University, London ON

June 2026

Bachelor of Engineering, Software Engineering | **Dean's Honour List**

- **Core Coursework:** Algorithms & Data Structures; Operating Systems; DBMS; Networking; Web Technologies; Information Security; Machine Learning; Cloud Computing; Software Testing & Maintenance.

Experience

Software Developer Intern, BroadGrain Commodities | Toronto, ON

Summer 2025

- **Rebuilt the company's compliance platform** as sole developer, consolidating a legacy .NET web app and MAUI mobile app into a unified cross-platform **React Native / TypeScript** interface.
- Engineered a **schema-driven backend factory** (Node.js) that auto-generates CRUD controllers, routes, and validation from a single config, reducing 100+ files to 3.
- Integrated **Microsoft Graph API + Azure AD** to automate SharePoint document delivery with token caching, retries, and exponential backoff.
- **Migrated all production data** to a redesigned normalized schema after resolving structural mismatches from the original denormalized database.
- Built **10 PDF reporting services** (JSReport, Handlebars, digital signatures, multi-photo support) to replace paper-based compliance workflows across 8 categories.
- **Reduced processing time by 15% through refactoring and 75% through digitization** across all reporting workflows.

Software Engineering Capstone, GTA Management System | Western University

2025–2026

- Built and **deployed a full-stack platform** (Next.js / TypeScript front end, Node.js / Express REST back end, MySQL) replacing spreadsheet-based GTA hiring at Western Engineering; **in active department-wide use across 100+ TAs, professors, and committee members**.
- Built the **React / TypeScript front end** with a reusable, fully-typed layout system (a generic dropdown shared across 6+ views) and a **drag-and-drop** interface for manual TA reassignment.
- Developed **78+ REST API endpoints** with role-based access control (student / professor / committee), Zod validation, and ownership enforcement; covered by an automated test suite.
- Designed a **configurable matching algorithm** combining skill match, preferences, course history, and review data with adjustable weights, executing assignments across 4 priority waves.
- Implemented a **dry-run assignment system** enabling preview, weight tuning, and manual overrides before any assignments are committed.

Projects

Soi Ram | Freelance Restaurant Website | soiram.ca

2025–2026

- Delivered a **production-grade Next.js / TypeScript / Tailwind site** with dynamic theming, fluid clamp() typography, and data-driven menu rendering; managed the full client lifecycle from design to deployment.
- Integrated **Toast POS ordering** alongside UberEats and DoorDash to enable commission-free direct orders with a unified receipt-style UI.
- Designed a **retro 80s brand identity** with custom typography, mobile-first layout, and SEO optimization.

Fracture Protocol | 2-Person Unity Roguelike

2025–2026

- Architected a **runtime modifier dispatch system** using string IDs queried at runtime, enabling dynamic effect composition across 32 modifiers (5 rarity tiers) without hardcoded branches.
- Implemented **coroutine-driven boss AI** for 3 encounters, running concurrent behaviors (fire patterns, drone phases, respawn timers) with phase transitions at 66%/33% HP; 77 scripts total in a 2-person Git workflow.

Skills

Languages & Frameworks: TypeScript/JavaScript, Java, C#, Python, SQL; React/React Native, Next.js, Node.js, Express.js, Tailwind CSS, Unity

Tools & Platforms: MySQL, Firestore, PostgreSQL, MongoDB; Prisma ORM; Azure/SharePoint, IIS, Nginx, Docker, Git, CI/CD; AI-assisted development (Claude Code); Networking (IP, Routing & Switching)

Spoken Languages: English, French